

Responsive Web design framework

Let us discuss some popular frameworks to create RWD, which look good and are easy to render to all devices.

Previously, RWD websites were created by writing a media query with the help of HTML and CSS. In a media query, we first add class in HTML based on the device's requirement. Then we mention the column, write breakpoints in, and add CSS in the CSS file with the help of `@media`.

For example: If the browser window is 600px or smaller, the background colour should be blue:

```
@media only screen and (max-width: 600px) {
  body {
    background-color: blue;
  }
}
```

When the screen (browser window) gets smaller than 768px, each column should have a width of 100 per cent:

```
/* For desktop: */
.col-1 {width: 8.33%;}
.col-2 {width: 16.66%;}
.col-3 {width: 25%;}
.col-4 {width: 33.33%;}
.col-5 {width: 41.66%;}
.col-6 {width: 50%;}
.col-7 {width: 58.33%;}
.col-8 {width: 66.66%;}
.col-9 {width: 75%;}
.col-10 {width: 83.33%;}
.col-11 {width: 91.66%;}
.col-12 {width: 100%;}

@media only screen and (max-width: 768px) {
  /* For mobile phones: */
  [class*="col-"] {
    width: 100%;
  }
}
```

Nowadays, a RWD framework can be easily downloaded and be added to the code, or be directly integrated as a cloud link or CDN link in the code. Thus, less code is written and the website looks good and responsive, as per current requirements.

Now, let's get acquainted with some popular RWD frameworks.

Bootstrap

Bootstrap is one of the most popular open source HTML, CSS and JS front-end frameworks. Its latest available version is 4.1.3. In Bootstrap, a 12-grid responsive layout is available, apart from 13 jQuery plugins, SCSS (which is

common for UI like carousals), pop-ups, modals, and other customised Bootstrap classes. Bootstrap is easy to learn and is used for creating responsive websites.

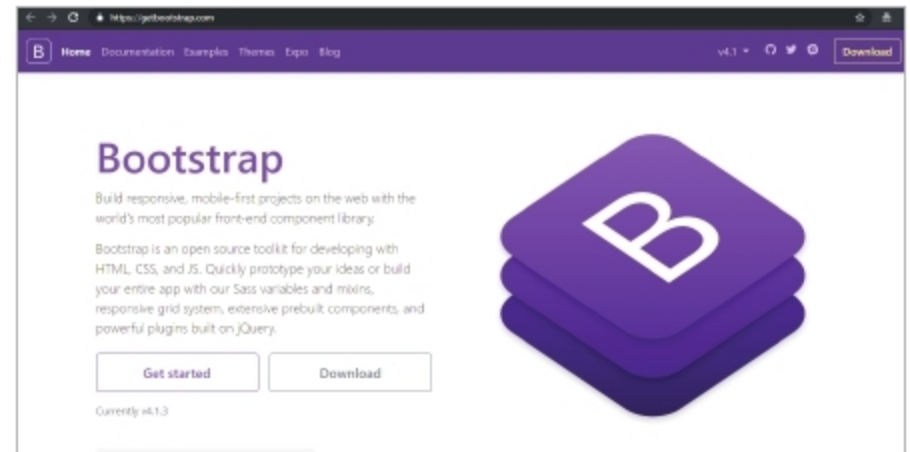


Figure 1: Bootstrap

There are two methods to install Bootstrap in your project.

1. Bootstrap package installation using npm requires you to use the following code:

```
npm install bootstrap // if we write only bootstrap then its
install latest
version of bootstrap
npm install bootstrap -v 3.0.0 (if you install specific
version of bootstrap)
```

2. *The CDN method:* When you are only using CSS and JS, you can use this method.

- i) For CSS, add the CDN link in your head section, as follows:

```
<link rel="stylesheet" ref="https://stackpath.bootstrapcdn.
com/bootstrap/4.1.3/css/bootstrap.min.css" integrity="sha384-
MCw98/ SFnGE8fJT3GXwEOngs V7Zt27NXFoaoApmYm81
iuXoPkF0JwJ8ERdknLPM0" crossorigin="anonymous">
```

- ii) For JS, Proper.js and jQuery, use the following code:

```
<script src="https://code.jquery.com/jquery-3.3.1.slim.min.
js" integrity="sha384-q8i/X+965Dz00rT7abK41JStQIAqVgRVzpbzo5sm
XKp4YfRvH+8abtTE1Pi6jizo" crossorigin="anonymous">
</script>
```

```
<script src="https://cdnjs.cloudflare.com/ajax/libs/popper.
js/1.14.3/umd/popper.min.js" integrity="sha384-ZMP7rVo3mIykV+2
+9J3UJ46jBk0WLaUAdn689aCwoqBjSiSnjAK/l8WvCWPIpM49" crossorigin
="anonymous">
</script>
```

```
<script src="https://stackpath.bootstrapcdn.com/
bootstrap/4.1.3/js/bootstrap.min.js" integrity="sha384-Chfqqxu
ZUCnJSK3+MXmPNIyE6ZbWh2IMqE241rYiqJxyMiZ6OW/JmZQ5stwEULTy" cro
ssorigin="anonymous">
</script>
```